

8 oct

08 October 2025 10:10

Pooled t-test

$N_p(\mu_1, \Sigma_1) \leftarrow \tilde{x}_1, \dots, \tilde{x}_{n_1}$ random sample $\begin{cases} \text{sample mean } \bar{X} \sim (p \times 1) \\ \text{sample cov } S_1 = \frac{1}{n_1-1} \sum_{i=1}^{n_1} (x_i - \bar{x})(x_i - \bar{x})^T \end{cases}$ $(p \times p)$

$N_p(\mu_2, \Sigma_2) \leftarrow \tilde{y}_1, \dots, \tilde{y}_{n_2}$ random sample $\begin{cases} \text{sample mean } \bar{Y} \sim (p \times 1) \\ \text{sample cov } S_2 \end{cases}$

each x_i is independent of y_j
no. of features is same but no. of samples may be dif.

$$\begin{aligned} H_0 : \mu_1 &= \mu_2 \\ H_1 : \mu_1 &\neq \mu_2 \end{aligned}$$

$E(\bar{X} - \bar{Y}) = \mu_1 - \mu_2$

$var(\bar{X} - \bar{Y}) = var(\bar{X}) + var(\bar{Y}) - 2cov(\bar{X}, \bar{Y})$ $\rightarrow 0$ (independent)

$= \frac{\Sigma_1}{n_1} + \frac{\Sigma_2}{n_2}$

case(1) : Σ_1 and Σ_2 are unknown but somehow we have this info that $\Sigma_1 = \Sigma_2 = \Sigma$

case(2) : Σ_1 and Σ_2 are unknown but we know that $\Sigma_1 \neq \Sigma_2$

case 1 $\Sigma_1 = \Sigma_2 = \Sigma$ (say) [Σ is unknown]

$var(\bar{X} - \bar{Y}) = \Sigma \left(\frac{1}{n_1} + \frac{1}{n_2} \right)$

Since we assume normal dist. $\bar{X} \sim N_p(\mu_1, \frac{\Sigma}{n_1})$, $\bar{Y} \sim N_p(\mu_2, \frac{\Sigma}{n_2})$

$\bar{X} - \bar{Y} \sim N_p(\mu_1 - \mu_2, \Sigma \left(\frac{1}{n_1} + \frac{1}{n_2} \right))$

$\Rightarrow \left(\frac{1}{n_1} + \frac{1}{n_2} \right)^{-1/2} (\bar{X} - \bar{Y} - (\mu_1 - \mu_2)) \sim N_p(0, \Sigma)$

under H_0 $\left(\frac{1}{n_1} + \frac{1}{n_2} \right)^{-1/2} (\bar{X} - \bar{Y}) \sim N_p(0, \Sigma)$

pooled :

$n_1 + n_2 \left\{ \begin{matrix} X \\ \bar{X} \\ Y \end{matrix} \right\}$ $\begin{matrix} \nearrow n_1 \times p \\ \nwarrow n_2 \times p \end{matrix}$

$$S_{pooled} = \frac{n_1-1}{n_1+n_2-2} S_1 + \frac{n_2-1}{n_1+n_2-2} S_2$$
$$(p \times p)$$

$E(S_{pooled}) = \frac{n_1-1}{n_1+n_2-2} E(S_1) + \frac{n_2-1}{n_1+n_2-2} E(S_2)$

$= \frac{n_1-1}{n_1+n_2-2} \Sigma + \frac{n_2-1}{n_1+n_2-2} \Sigma = \Sigma$

why n_1+n_2-2

$\therefore S_{pooled}$ is unbiased estimator of Σ

$(n_1-1) S_1 \rightarrow$ Wishart $W_p(n_1-1, \Sigma)$ = sum of n_1-1 indepent 1 rank matrix

$\uparrow$
df

$(n_1-1) S_1 + (n_2-1) S_2 \rightarrow$ sum of n_1+n_2-2 independent 1 rank matrix [additive bcoz same Σ]

$(n_1+n_2-2) S = (n_1-1) S_1 + (n_2-1) S_2 \sim W_p(n_1+n_2-2, \Sigma)$

$T^2 = n(\bar{X} - \mu)^T S^{-1} (\bar{X} - \mu)$

re-write it as

$= \sqrt{n} (\bar{X} - \mu)^T \left(\frac{(n-1)S}{n-1} \right)^{-1} \sqrt{n} (\bar{X} - \mu)$

$\rightarrow N_p(0, \Sigma)$ $\rightarrow W_p(n-1, \Sigma)$

$$\frac{\bar{x} - \mu}{1/\sqrt{n}} \sim N_p(0, \Sigma)$$

Note the pattern in T^2

$$T^2 = \left(\frac{1}{n_1} + \frac{1}{n_2} \right)^{-1/2} \left(\bar{\tilde{x}} - \bar{\tilde{y}} - (\mu_1 - \mu_2) \right)^T \left[\frac{(n_1-1)S_1 + (n_2-1)S_2}{n_1+n_2-2} \right]^{-1} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)^{-1/2} \left(\bar{\tilde{x}} - \bar{\tilde{y}} - (\mu_1 - \mu_2) \right)$$

$$= \left(\frac{1}{n_1} + \frac{1}{n_2} \right)^{-1} \left(\bar{\tilde{x}} - \bar{\tilde{y}} - (\mu_1 - \mu_2) \right)^T S_{\text{pooled}}^{-1} \left(\bar{\tilde{x}} - \bar{\tilde{y}} - (\mu_1 - \mu_2) \right)$$

under H_0 :

$$T^2 = \left(\frac{1}{n_1} + \frac{1}{n_2} \right)^{-1} \left(\bar{\tilde{x}} - \bar{\tilde{y}} \right)^T S_{\text{pooled}}^{-1} \left(\bar{\tilde{x}} - \bar{\tilde{y}} \right)$$

recall $\frac{(n-1)p}{n-p} F_{p, n-p, \alpha}$

compared

In our context $\rightarrow \frac{p(n_1+n_2-2)}{n_1+n_2-1-p} F_{p, n_1+n_2-1-p, \alpha} \rightarrow \textcircled{3}$

if $T^2 > \textcircled{3}$ reject H_0

Case ② $\Sigma_1 \neq \Sigma_2$

$$T^2 = (\bar{x} - \bar{y} - (\mu_1 - \mu_2))^T \left(\frac{S_1}{n_1} + \frac{S_2}{n_2} \right)^{-1} (\bar{x} - \bar{y} - (\mu_1 - \mu_2))$$

approximately $\frac{\nu p}{\nu - p + 1} F_{p, \nu - p + 1, \alpha}$ distributed

$$\nu = \left\lceil \frac{(\text{trace}(A^2))^2}{\text{trace}(A^4)} \right\rceil ; A_{p \times p} = \left(\frac{S_1}{n_1} + \frac{S_2}{n_2} \right) \left(\frac{S_1^2}{n_1^2(n_1-1)} + \frac{S_2^2}{n_2^2(n_2-1)} \right)$$